World Models in Action

3. Representation learning before generation

This chapter covers

- How self-supervised objectives define useful invariances
- Contrastive learning, bootstrapping, clustering and redundancy reduction
- Masked prediction across visual, language and speech representations
- How to inspect a learned representation without overclaiming

A world model needs an internal state that preserves distinctions relevant to predicting what happens next. Raw observations are often too high-dimensional and entangled to serve directly as a convenient state. An encoder transforms them into a representation, but the transformation cannot be judged by compression alone. It must retain the distinctions that a dynamics model or planner will need.

When labels are unavailable, the encoder still needs a learning signal. Self-supervised learning obtains that signal from the data itself: two views can be treated as related, one representation can predict another, feature statistics can be constrained, or missing content can be predicted from context. These procedures differ in what they reward the representation for retaining.

The central idea is that an objective is not a neutral way to optimize an encoder. It specifies a relationship the model must satisfy. The way that relationship is constructed determines which variations are ignored, which distinctions are amplified, and which information may be lost. No objective guarantees that its representation is a sufficient physical state. That claim requires separate evidence.

The methods in this chapter are arranged by the problem each family addresses, not by publication year. We derive one contrastive objective fully, compare strategies for avoiding collapsed features, and examine how masked prediction changes when the target is a pixel, a token or another model's representation. The examples explain learning mechanisms; they are not a benchmark ranking of methods trained with different data, architectures and computational budgets.

3.1. The objective is a contract

3.1.1. A representation is not just a compressed observation

Let x be an observation and let f_θ be an encoder with parameters θ. Its output is a vector z:

$$z = f_\theta(x).$$

A smaller representation is useful only if the information removed is unnecessary for the task. A compact vector might preserve object identity but lose velocity. It might preserve average appearance but erase the visual cue that distinguishes an open gripper from a closed one. Compression creates the possibility of a useful state and the possibility of aliasing.

Chapter 2 formalized the desired contract in terms of sufficiency, minimality and aliasing. Self-supervised objectives do not observe that contract directly. They rely on a proxy relationship in the data. A representation may perform well on the proxy task while failing to retain information the world model needs.

3.1.2. Invariance is a decision

Suppose T_1 and T_2 are random transformations applied to the same input:

$$x_1 = T_1(x), \qquad x_2 = T_2(x).$$

A view-consistency objective encourages $f_\theta(x_1)$ and $f_\theta(x_2)$ to agree. The design decision is hidden in T_1 and T_2: they define which input differences the model is encouraged to ignore.

Random crops are useful when the cropped-out region is irrelevant to the intended representation. They can be harmful when the missing region contains a control-relevant object or a velocity cue. Color jitter can suppress nuisance lighting changes, but it can also remove a visual marker used for localization. The transformation is part of the model's inductive bias, not a neutral augmentation step.

An ideal invariance would identify observations that the downstream task truly treats as equivalent. A practical augmentation only approximates that relation. We should not assume that two views are interchangeable for every task just because they came from the same image.

Historically, visual self-supervision used hand-designed pretext tasks such as predicting image rotation (Gidaris et al., 2018), restoring the spatial order of jigsaw patches (Noroozi and Favaro, 2016), predicting spatial context (Doersch et al., 2015), and colorizing grayscale images (Zhang et al., 2016). These tasks assign synthetic targets to unlabeled data and make the target-design assumption easy to see. They remain useful conceptual examples, but the architecture survey below focuses on the broader contrastive, bootstrap, statistical, clustering and masked-prediction families that followed.

The principle behind representation learning

Every self-supervised method constructs a target relationship from the data. The relationship may be agreement between views, a ranking among candidates, a teacher's output distribution, statistics of a feature batch, a cluster assignment or a missing patch. Ask what information makes that relationship easy to satisfy, then ask whether that information is

what the downstream model needs.

figure 3.1 maps the major objective families by their dominant training signal; the map deliberately allows overlap between methods.

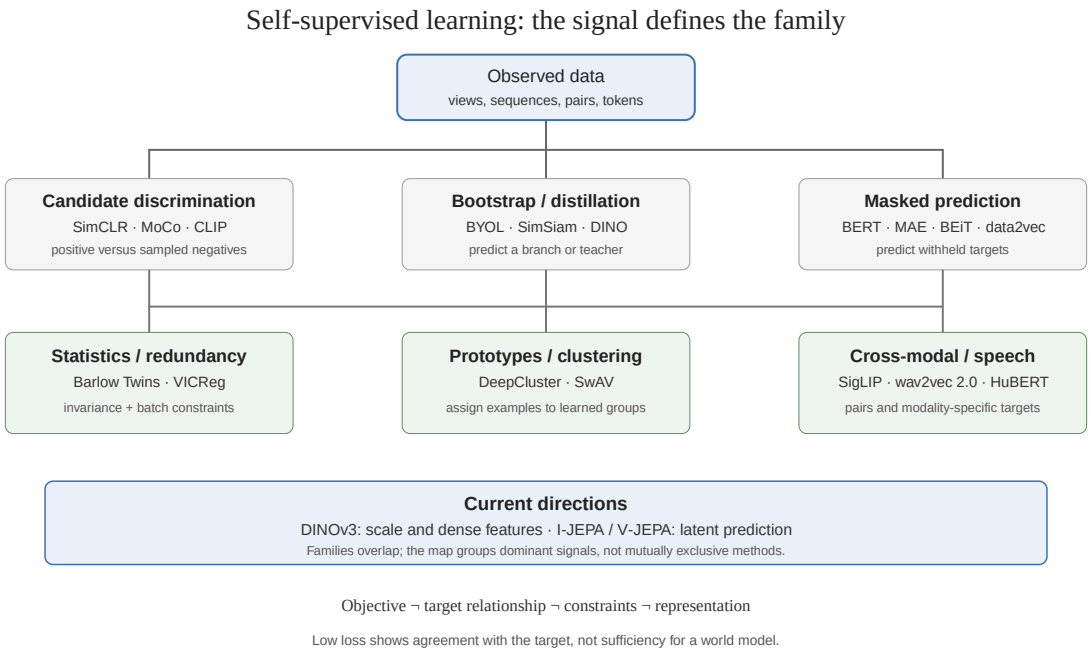


Figure 3.1 The major families of self-supervised learning

The map is not a partition into disjoint categories. DINO uses teacher-student self-distillation over prototype outputs; iBOT joins masking and teacher targets; wav2vec 2.0 combines masking, quantization and contrastive discrimination.

3.2. Contrastive learning

3.2.1. From pairs to candidate classification

A contrastive method begins with an anchor, a related positive example and distractors called negatives. The encoder maps each example to a feature vector. A similarity function scores the anchor against each candidate. Training rewards the positive for receiving a higher score than the negatives.

For two feature vectors, cosine similarity is their dot product after unit normalization:

$$s(z_i, z_j) = \frac{z_i^\top z_j}{\| z_i \|_2 \, \| z_j \|_2}.$$

A temperature $\tau>0$ scales the similarity before applying softmax. A lower temperature makes the candidate distribution sharper and increases the effect of high-scoring candidates. It also changes gradient scale and optimization sensitivity.

Suppose a batch contains N source images. Two stochastic views of each image give $2N$ encoded vectors. For anchor i, let $p(i)$ be the index of its other view. Excluding the anchor itself, every remaining view is a candidate. The NT-Xent loss for that anchor is

$$\ell_i = -\log \frac{\exp(s(z_i, z_{p(i)}) / \tau)}{\sum_{a=1}^{2N} \mathbf{1}[a \neq i] \exp(s(z_i, z_a) / \tau)}.$$

The total loss averages over the anchors. In the denominator are one positive and $2N-2$ negatives. The positive is included in the denominator; the anchor is not. Stating the candidate set explicitly avoids ambiguity because papers do not all use the same symbol for the number of candidates or negatives.

This objective is ordinary softmax cross-entropy. Write $a_{ij} = s(z_i, z_j) / \tau$, and let q_{ij} be the softmax probability assigned to candidate j. Differentiating with respect to each candidate logit gives

$$\frac{\partial \ell_i}{\partial a_{ij}} = q_{ij} - \mathbf{1}[j = p(i)].$$

The positive logit receives gradient $q_{i,\,p(i)} - 1$, which pushes it upward under gradient descent. A negative logit receives gradient q_{ij}, which pushes it downward. The gradient on a negative is weighted by its current probability: a hard negative receives more repulsive gradient than an already dissimilar candidate. Since $a_{ij} = s(z_i, z_j) / \tau$, differentiation with respect to similarity introduces the factor $1 / \tau$.

figure 3.2 connects the positive-pair construction to the candidate distribution and its gradient.

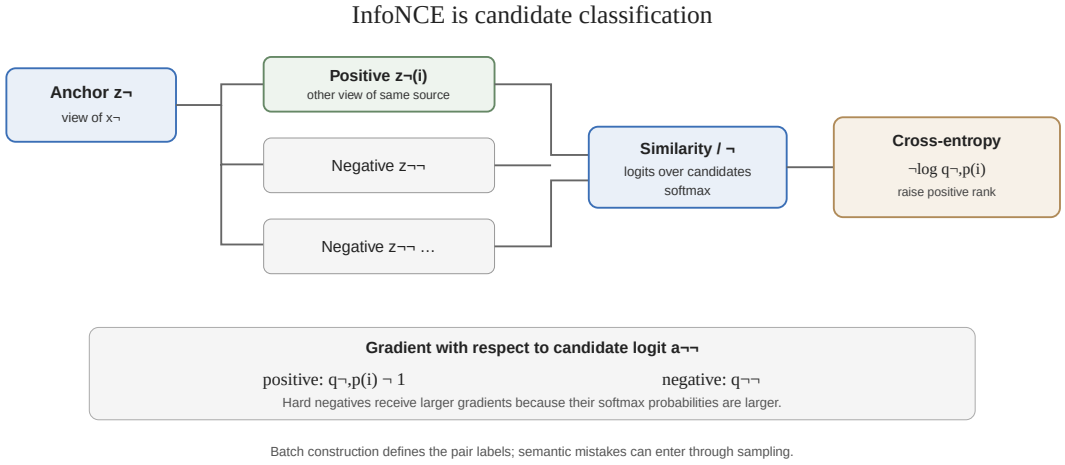


Figure 3.2 One anchor, one positive and a candidate distribution

3.2.2. What InfoNCE does and does not prove

The denominator makes the objective relative: the model must distinguish the designated positive from sampled competitors. Maximizing positive similarity alone would not impose that ranking constraint. With unnormalized dot products it can also reward increasing feature norms; unit normalization bounds the score but does not remove dependence on the sampling protocol.

The labels “positive” and “negative” come from the data construction. A negative may be an image of the same semantic category as the anchor. This is called a false negative relative to that semantic task. It is not an algebraic error: it is a mismatch between the sampling label and the relationship we want the representation to encode. Augmentations create a related risk. If a transformation destroys a relevant feature, the loss may teach the encoder to ignore that feature.

InfoNCE is often connected to mutual information. For a specified joint distribution of positive pairs and an appropriate negative-sampling distribution, the optimal critic is related to the density ratio between joint and independent samples. Under those assumptions, a commonly used form of the bound is

$$I(X;Y) \geq \log K - \mathcal{L}_{\mathrm{NCE}},$$

where K denotes the number of candidates under the convention used by the derivation. Candidate-sampling assumptions are essential. A loss called InfoNCE does not automatically produce this bound under every arbitrary negative-sampling scheme.

The bound also has a ceiling of $\log K$ for finite candidate count. Increasing mutual information beyond that ceiling cannot tighten this particular bound without increasing K. More fundamentally, maximizing dependence between views is not identical to retaining information needed for a downstream physical prediction. The chapter therefore derives the classification objective and its gradient first; the information-theoretic interpretation is useful, but it is not a universal guarantee of representation quality.

3.2.3. SimCLR and MoCo: two ways to construct candidates

SimCLR (Chen et al., 2020) made the augmented-view recipe explicit. Two views pass through an encoder f and a nonlinear projection head g. The contrastive objective is applied to the projected vectors, while the encoder representation is typically what downstream tasks use. That distinction matters: the loss space and the feature space are not necessarily the same. SimCLR’s results also depended on augmentation composition, batch size, architecture, training duration and optimization. Its contribution is a clear and strong recipe, not a proof that every contrastive method needs one specific batch size or projection head.

MoCo (He et al., 2020) addressed a different engineering issue: keeping a large candidate dictionary without requiring every negative to come from the current batch. It uses a query encoder

and a key encoder. The key encoder follows an exponential moving average:

$$\theta_k \leftarrow m\theta_k + (1-m)\theta_q.$$

Encoded keys are stored in a queue. A query can therefore compare with many keys produced by earlier minibatches. Since queued keys become stale as parameters change, the slowly moving key encoder reduces, but does not eliminate, dictionary inconsistency. MoCo v2 (Chen et al., 2020) strengthens the training recipe; MoCo v3 (Chen et al., 2021) studies a changed recipe for vision transformers. These developments should be described with their actual architecture and training context, not as arbitrary version numbers.

Both methods remain sensitive to the definition of positive pairs and negatives. A queue solves the size and update-consistency problem; it does not establish that every negative is semantically different.

3.2.4. CPC as a historical bridge

Contrastive Predictive Coding (CPC; van den Oord et al., 2018) encodes a sequence and uses a context vector from the past to score future latent representations. An InfoNCE-style objective teaches the context to identify the observed future latent among candidate latents. CPC is a useful historical bridge between contrastive discrimination and predictive representations, but it is not the centerpiece of a contemporary visual SSL survey.

A CPC future-latent retrieval score is not the same as an action-conditioned dynamics model. If the predictor has no action input, it does not specify which action caused the future. Successful retrieval does not establish recursive physical rollout accuracy, Markov sufficiency or planning value. Those are separate claims requiring separate tests.

3.3. Learning without explicit negatives

Contrastive learning supplies an explicit competitor set. Another family removes that dictionary and trains one branch to predict a target produced by another branch. This avoids explicit negative pairs, but it changes rather than eliminates the collapse question.

3.3.1. BYOL and SimSiam: bootstrap prediction

BYOL (Grill et al., 2020) uses an online encoder and projection head, an extra predictor on the online branch, and a target encoder with a target projection head. The target branch is stop-gradient, and its parameters are an exponential moving average of the online parameters. A normalized regression loss asks the online prediction to match the target representation of another view. Training can also swap the roles of the two views.

SimSiam (Chen and He, 2021) keeps the predictor and stop-gradient asymmetry but removes the

moving-average target update. Both branches share encoder parameters; the detached target branch does not receive gradient through that loss term. The important operation is not just two encoders that agree. It is which pathway receives gradients and how the predictor and normalizations affect parameter updates.

A constant representation can make a simple view-agreement loss zero. Thus neither predictor asymmetry nor EMA should be presented as a standalone mathematical theorem that collapse is impossible in all settings. A collapsed solution may exist yet be unstable under a particular training dynamics; an empirical model may avoid collapse for a particular recipe without a universal proof. Distinguish the existence of a collapsed solution, its stability, and the outcome of a particular run.

3.3.2. DINO: match a teacher's distribution

DINO (Caron et al., 2021) replaces direct feature regression with distribution matching. A student and a teacher map different views to probability distributions over learned prototype outputs. The student is trained by cross-entropy to match a teacher distribution that has been centered and sharpened. The teacher parameters follow an EMA of the student parameters.

Centering and sharpening have different roles: centering modifies output statistics to resist concentration on a small set of prototypes; sharpening lowers the entropy of the teacher's target distribution. Prototype indices are learned output coordinates, not automatically human-defined categories. DINO is best understood as self-distillation with a prototype distribution, rather than as an ordinary clustering algorithm.

DINOv2 (Oquab et al., 2023) and DINOv3 (Siméoni et al., 2025) demonstrate that representation quality depends on more than the named loss. Data curation, model and data scale, architecture, resolution, schedule and training diagnostics are all part of the system. DINOv3, introduced in 2025, emphasizes scalable dense features and introduces Gram anchoring to address degradation of feature maps during extended training. That development is a reminder that global image representations and high-quality local features are different objectives to monitor.

3.4. Redundancy reduction and explicit variance constraints

BYOL, SimSiam and DINO make training dynamics or teacher targets central to avoiding trivial solutions. Barlow Twins (Zbontar et al., 2021) and VICReg (Bardes et al., 2022) express another idea: require paired views to agree, but constrain the statistics of features across a batch.

3.4.1. Barlow Twins: cross-correlation toward identity

Let Z^A and Z^B be feature batches from two views. Normalize each feature dimension across the batch and define cross-correlation C:

$$C_{ij} = \frac{1}{N} \sum_{b=1}^{N} \tilde{z}^{A}_{b,\,i} \tilde{z}^{B}_{b,\,j}.$$

If corresponding dimensions agree and distinct dimensions are uncorrelated across views, C approaches the identity matrix. Barlow Twins minimizes the diagonal mismatch and penalizes off-diagonal cross-correlation:

$$L_{\mathrm{BT}} = \sum_{i} (1 - C_{ii})^2 + \lambda \sum_{i} \sum_{j \neq i} C_{ij}^2.$$

The diagonal terms align a feature with itself across views; the off-diagonal terms discourage redundant correlations between different dimensions. This is a second-order statistical objective. An identity cross-correlation matrix does not prove that features are statistically independent under arbitrary distributions, and it does not prove that each coordinate has a useful physical interpretation.

3.4.2. VICReg: make three constraints explicit

VICReg separates invariance, variance and covariance. For two feature batches Z and Z', the invariance term is a mean-squared difference:

$$s(Z, Z') = \frac{1}{Nd} \| Z - Z' \|_F^2 .$$

For each feature dimension j, let $\sigma_j(Z)$ be its standard deviation over the batch. A variance penalty puts a floor under each dimension's standard deviation:

$$v(Z) = \frac{1}{d} \sum_{j} \max(0, \gamma - \sigma_j(Z)).$$

The covariance term penalizes the squared off-diagonal entries of the empirical covariance matrix:

$$c(Z) = \frac{1}{d} \sum_{i \neq j} (Z)_{ij}^2.$$

The full objective is a weighted sum of the invariance term and the corresponding variance and covariance terms for both views. The published implementation's coefficients and numerical choices matter; the equations state the structural roles.

If every example has the same value in one dimension, its batch standard deviation is zero and the variance penalty is positive for any positive margin. That makes the anti-collapse constraint explicit. It does not prove that the surviving variation is semantically relevant. A feature can vary with background texture and still satisfy a variance floor.

The methods are related but not identical. Barlow Twins uses normalized cross-correlation as the central matrix objective. VICReg uses a direct invariance term, a standard-deviation floor and a covariance penalty. Their shared lesson is that view agreement and non-collapse are separate requirements. Their moment constraints are not a guarantee of a world-model-ready state.

figure 3.3 compares the target path and anti-collapse mechanism used by the major non-identical objective families.

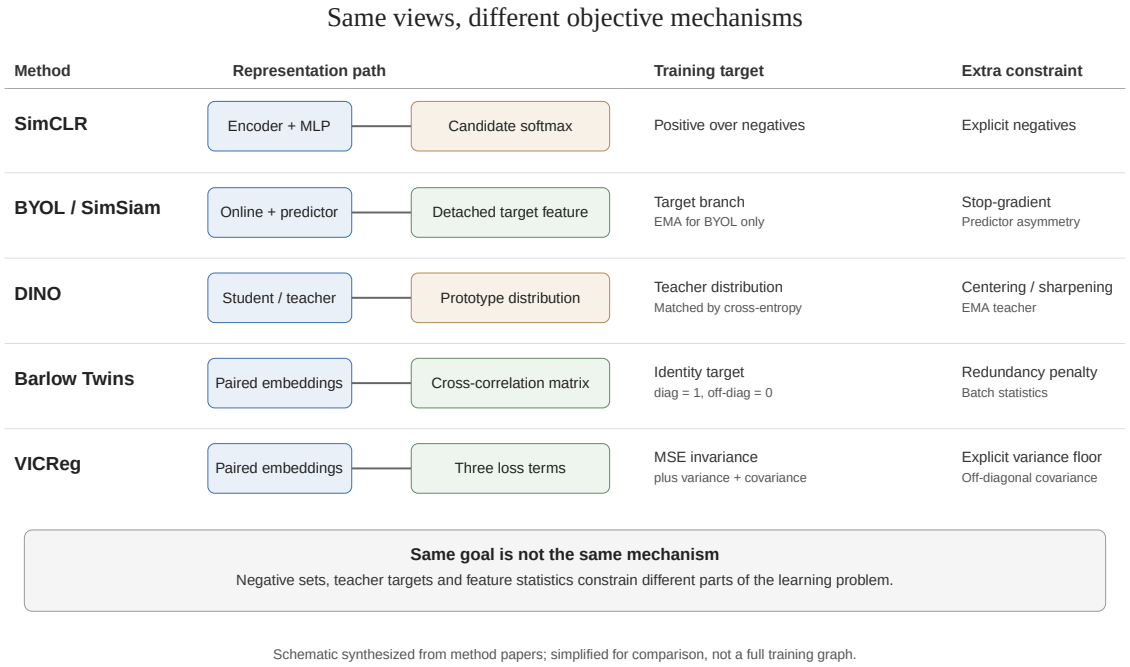


Figure 3.3 Different methods make different target paths and constraints explicit

3.5. Clusters and prototypes as learning targets

Contrastive methods make instances compete; clustering methods instead construct pseudo-categories from the feature space. Their premise is that examples may share a prototype-level relation rather than only match as exact instances.

DeepCluster (Caron et al., 2018) alternates feature extraction, k-means assignments and network training to predict the current cluster IDs. This provides pseudo-labels without human annotation, but assignments depend on current features and can be stale or unbalanced. Cluster identities are arbitrary; permuting all cluster labels leaves the partition unchanged.

SwAV (Caron et al., 2020) moves assignment online. Learned prototypes assign examples to clusters, and the network predicts the assignment of one view from another view. A Sinkhorn-Knopp procedure approximately balances assignments across prototypes. Balancing discourages collapse to a single prototype, but it also encodes an assumption: useful prototype usage should be

spread across the batch. A truly skewed dataset may not satisfy that assumption.

DINO also predicts distributions over prototypes, but its central training description is teacher-student self-distillation. This overlap is useful: method taxonomies describe dominant mechanisms, not exclusive boxes. The essential questions are how targets are assigned, whether assignments are fixed or online, how imbalance is handled, and what downstream evidence validates the clusters.

3.6. Masked prediction: choose what the model must recover

3.6.1. One mask, several target spaces

Suppose an input is divided into visible positions V and masked positions M. An encoder processes the allowed context and a prediction head estimates a target y_i at masked positions:

$$L = \frac{1}{|M|} \sum_{i \in M} \ell(\hat{y}_i, y_i).$$

This form leaves the crucial choice open: what is the target? It might be pixel values, a discrete code, a word token, a replaced-token label or a contextualized feature. The mask tells the model where information is withheld; the target tells it what relationship must be learned.

figure 3.4 compares three target spaces for masked inputs: pixels, discrete visual codes and contextualized features.

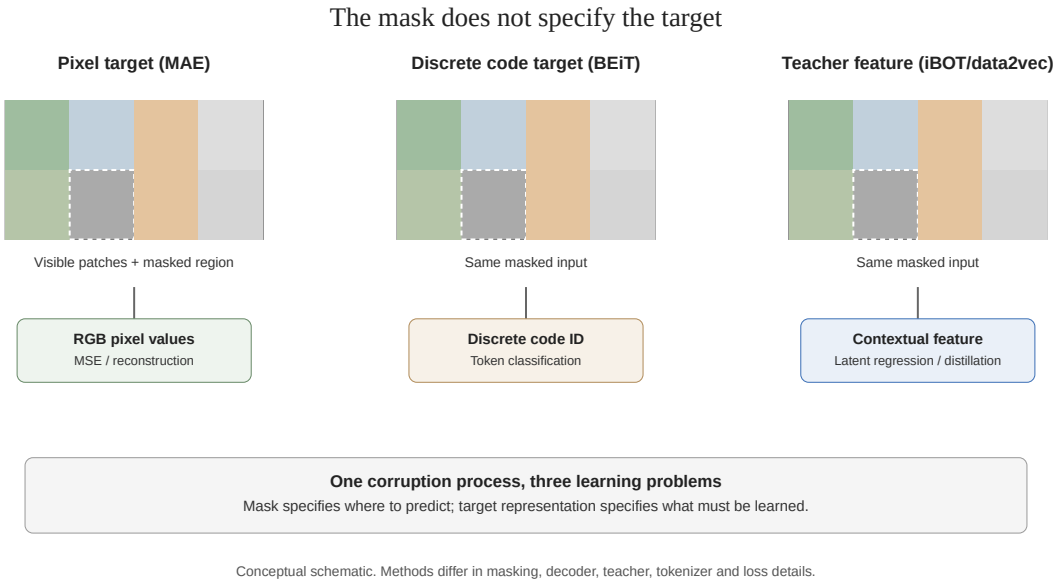


Figure 3.4 The target space changes the prediction problem

3.6.2. MAE: predict masked pixels

A masked autoencoder (MAE; He et al., 2022) hides a large fraction of image patches and sends only visible patches through the encoder. A smaller decoder receives the encoded visible patches plus mask tokens and predicts patch pixels; the loss is applied to masked patches. This asymmetric design reduces encoder computation because masked patches do not pass through the expensive encoder.

Under squared error, the conditional mean minimizes expected prediction risk. When several patch completions are plausible from visible context, that mean can average them. The limitation is not that pixel reconstruction is always poor; it is that reconstructing pixels does not guarantee the encoder preserves the state variables a downstream model needs. The encoder may be retained for transfer while the reconstruction decoder is discarded.

3.6.3. BEiT, iBOT and data2vec: change the target

BEiT (Bao et al., 2022) predicts discrete visual-token IDs produced by a learned tokenizer. Rather than regress pixel values, the model classifies the tokenizer's code for each masked patch. The tokenizer introduces its own prior: useful structure can be learned from the code vocabulary, but tokenizer errors and biases enter the target.

iBOT (Zhou et al., 2022) combines masked patch prediction with teacher-student self-distillation. It is an example where a family label is insufficient: the input is masked, target distributions come from a teacher, and prototype outputs structure those targets. Other masked-image methods deliberately change the target: SimMIM (Xie et al., 2022) uses a simple pixel-reconstruction objective; MaskFeat (Wei et al., 2022) predicts hand-designed visual features such as HOG rather than raw pixels; BEiT v2 improves the discrete-target route with semantic visual tokenization. These variations reinforce the same principle: the mask alone does not define the learning task.

data2vec (Baevski et al., 2022) predicts contextualized teacher features from masked input and was designed as a framework spanning speech, vision and language. A latent target may contain broader context than a local pixel target, but matching a teacher does not automatically ensure that the feature preserves all future-relevant or causal state information. In every case, the representation is shaped by the selected target.

3.6.4. BERT, GPT and ELECTRA: self-supervision in language

BERT (Devlin et al., 2019) uses masked language modeling: a subset of tokens is hidden or corrupted, and the model predicts the original token identities using bidirectional context. GPT (Radford et al., 2019) uses causal next-token prediction. The chain rule factorizes sequence likelihood as

$$p(x_1, \ldots, x_T) = \prod_{t=1}^{T} p(x_t \mid x_{<t}).$$

Its context contains preceding tokens only. Both are self-supervised pretraining, but GPT is not a masked-language-model objective. This distinction illustrates the general principle that target and context are separate design choices.

ELECTRA (Clark et al., 2020) corrupts selected tokens and trains a discriminator to identify whether each input token was replaced. Its learning signal can apply at many positions, rather than only asking the model to reconstruct selected token IDs. These language methods are included to make the target-design principle general; the main visual architecture progression remains the focus here.

3.7. The same questions across modalities

CLIP (Radford et al., 2021) aligns image and text embeddings using a symmetric contrastive objective. In a batch of paired examples, each image is trained to retrieve its corresponding text and each text its corresponding image. The positive relation comes from captions or text associated with images. Such pairs provide semantic supervision, but captions may be incomplete, generic or noisy. Retrieval alignment does not guarantee detailed spatial grounding or physical understanding.

SigLIP (Zhai et al., 2023) replaces the batchwise softmax with pairwise sigmoid-style losses. This changes the normalization and scaling behavior, not the underlying need for reliable pair construction. SigLIP 2 (Tschannen et al., 2025) extends the family toward multilingual understanding, localization and dense features through a combined recipe that includes captioning-based pretraining, self-distillation, masked prediction and online data curation. Do not describe SigLIP as merely a temperature variant of CLIP.

Speech methods expose other target-construction choices. wav2vec 2.0 (Baevski et al., 2020) masks latent speech frames and contrasts contextual representations against quantized targets. HuBERT (Hsu et al., 2021) predicts cluster-derived pseudo-labels for masked speech regions. data2vec predicts contextual teacher features. table 3.1 compares methods by target, loss, signal construction and limitation rather than by a single benchmark score.

Table 3.1 Self-supervised objectives compared by the signal they construct

Method	Target	Loss family	What creates the signal	Important limitation
SimCLR	Matching view among candidates	Contrastive cross-entropy	Two augmented views and in-batch negatives	False negatives and augmentation assumptions
MoCo	Matching key among candidates	Contrastive cross-entropy	Momentum key encoder and queue	Queued keys can be stale
BYOL / SimSiam	Other branch's feature	Normalized feature regression	Predictor, target/stop-gradient asymmetry	Collapse is not excluded by agreement alone
DINO	Teacher distribution over prototypes	Distillation cross-entropy	EMA teacher, centering and sharpening	Feature structure is empirical, not guaranteed
Barlow Twins	Cross-view identity correlation	Squared correlation error	Batch cross-correlation	Second-order decorrelation is not semantic sufficiency
VICReg	Matching features plus moment constraints	MSE + variance + covariance	Explicit statistics of each view batch	Variation can encode nuisance information
DeepCluster / SwAV	Pseudo-cluster assignment	Cluster classification / assignment prediction	K-means or balanced online prototypes	Cluster balance and quality are assumptions
MAE	Masked pixel values	Reconstruction regression	Visible patches plus masked positions	Pixel reconstruction does not imply state sufficiency
BEiT / iBOT	Discrete codes or teacher patch targets	Token or distillation cross-entropy	Tokenizer / teacher outputs	Target bias is inherited

Method	Target	Loss family	What creates the signal	Important limitation
BERT / GPT	Masked token / next token	Token cross-entropy	Corrupted text or observed sequence order	Different context restrictions; not visual state objectives
CLIP / SigLIP	Paired image-text relation	Contrastive softmax / sigmoid	Natural language-image pairings	Alignment does not establish grounding
wav2vec 2.0 / HuBERT	Quantized or cluster-derived speech units	Contrastive / masked classification	Audio itself and learned pseudo-targets	Units depend on quantizer or clustering
data2vec	Contextual teacher representation	Latent regression	EMA teacher on less-corrupted input	Teacher targets may omit task-critical information

3.8. The current frontier and the boundary of this chapter

DINOv3 illustrates the recent effort to scale self-distilled visual features while preserving dense spatial information. The method's reported Gram anchoring addresses degradation of dense features during long training schedules. This is not just DINO with more parameters: data curation, architecture, optimization and feature diagnostics contribute to the system.

Joint-embedding predictive architectures are another important direction. I-JEPA (Assran et al., 2023) predicts representations of target image regions from a context representation. V-JEPA (Bardes et al., 2024) moves the target into video, and V-JEPA 2 (Assran et al., 2025) explores video representation learning for visual understanding and anticipation. V-JEPA 2-AC adds action-conditioned post-training for robotic control. These are distinct stages. Passive video pretraining alone does not specify how the world changes under an action.

As of October 2026, the official V-JEPA repository also lists V-JEPA 2.1 (Mur-Labadia et al., 2026), which adds a dense predictive loss over visible and masked tokens, self-supervision at multiple encoder depths and multimodal tokenizers. This is a refinement of latent prediction aimed at stronger spatially detailed features.

The JEPA family marks a boundary between learning an encoder from views or missing content and explicitly predicting target representations across space and time. Its architectural details

warrant a dedicated treatment. Here the main lesson is that representation objectives are becoming increasingly predictive and target-space-aware, while action-conditioned dynamics remains a separate question.

Diffusion and flow-matching objectives also train on unlabeled data, but their main purpose is to learn a generative process and sample new observations. They should not be collapsed into the contrastive, bootstrap, clustering and masked-feature families simply because training data lack human class labels. Their derivation belongs with generative modeling.

3.9. Two small demonstrations

The code companion contains two focused demonstrations. They support the derivations rather than serving as the chapter's main argument.

3.9.1. Demonstration 1: compute an InfoNCE batch

Use the small contrastive-loss implementation in ch03/03_ntxent_from_scratch.py. Trace one anchor through the following steps: identify its positive index; compute similarity logits; divide by temperature; exclude the anchor from the candidate denominator; apply cross-entropy; and inspect the gradient for positive and negative logits. Start with a tiny set of fixed vectors before adding an encoder. The test is successful when the implementation matches the equation and candidate count, not when a benchmark score is reached.

3.9.2. Demonstration 2: invariance is not enough

Use the small collapse illustration in ch03/06_collapse_demo.py to compare view invariance with explicit variance/covariance terms. Begin with constant vectors, for which agreement loss is zero. Then calculate why the VICReg variance term is nonzero when a dimension's standard deviation is below its margin. Treat this as an algebraic diagnostic of the stated objectives, not as a reproduction of the training scale or performance claims of BYOL, SimSiam or VICReg.

3.10. Auditing a representation for a world model

A low self-supervised loss answers whether the representation satisfies its training objective. It does not answer whether the representation is a useful state. A linear probe can reveal whether selected variables are linearly decodable, but cannot certify that all future-relevant information is present. A reconstruction score can show whether target pixels are predicted, but cannot certify causal state. A future-latent retrieval score tests candidate identification under the chosen sampling protocol, not necessarily physical rollout or action-conditioned prediction.

Use a sequence of checks whose claims are explicit:

1. **Inspect the view transformations.** Identify what information may be erased by crops, color changes, masking or temporal sampling.
2. **Inspect the target.** Determine whether it is a candidate ID, teacher feature, statistic, cluster assignment, token or pixel. Ask which information lowers its loss.
3. **Check collapse directly.** Monitor per-dimension standard deviations, covariance/correlation structure, feature rank and distribution across examples. A scalar training loss is not enough.
4. **Test relevant variables.** Use held-out labels or simulator state only for evaluation; state clearly which variables and data splits the probe uses.
5. **Test the distinction required by prediction.** For world models, include occlusion, velocity, changing geometry and relevant context where possible. Distinguish single-frame representation quality from sequence prediction.
6. **Test action conditioning and downstream use separately.** A model that predicts a passive future distribution need not know which future follows a selected action. Dynamics and planning require their own tests.

Three components, three different jobs

A static visual encoder maps a frame, or a fixed stack of frames, to a latent vector. An optional recurrent wrapper carries a learned state across observations. Neither is a complete world model. A full world model additionally needs a transition model that predicts future state or observations conditioned on actions, and a method for evaluating or planning over candidate futures.

The correct conclusion is narrower than a model-card claim: a method learns the information that makes its constructed target relationship easier to satisfy, subject to architecture and optimization. Whether that information is useful for a particular agent is a downstream question.

3.11. Summary

Self-supervised learning constructs a learning signal from data. Contrastive objectives turn positive-pair matching into candidate classification. SimCLR and MoCo show different ways to build the candidate set. BYOL, SimSiam and DINO use prediction or teacher distributions without explicit negative pairs, but agreement alone is compatible with collapse. Barlow Twins and VICReg make redundancy and variance constraints explicit. DeepCluster and SwAV use pseudo-labels or prototypes. Masked prediction shows that the target can be a pixel, discrete code, word, replaced-token label or contextualized feature.

Across methods, the same questions recur: which inputs are treated as equivalent, what target is

predicted, which context is available, how the objective handles degenerate solutions, and what evaluation supports the intended claim. Modern methods improve data, targets, architecture and training recipes in different ways; no single loss is a universal guarantee of a sufficient physical state.

For world modeling, representation learning is only one part of the system. A good state must retain distinctions that matter for future prediction. The encoder's self-supervised loss must not be confused with an action-conditioned dynamics objective or evidence of planning utility.

3.12. Exercises

1. Derive the gradient of softmax cross-entropy with respect to a positive logit and a negative logit. Explain why high-probability negatives receive larger gradients.
2. A minibatch uses two views of N source images. How many candidates enter each anchor's denominator in NT-Xent, and how many are negatives? Explain why stating the candidate set is safer than relying on notation alone.
3. Construct a world-model example in which crop invariance removes task-relevant information. Describe a second augmentation that is likely to help and a task where it may be harmful.
4. Explain why a constant representation minimizes pure view-invariance MSE. Which term in VICReg penalizes the same solution, and why does this not prove that the learned features are semantically useful?
5. Compare Barlow Twins and VICReg in terms of invariance, variance and redundancy constraints. Which quantities are estimated from the batch?
6. Explain the difference between BERT's masked-token objective and GPT's causal next-token objective. Which context can each model use to predict a token?
7. Give an example of a false negative in contrastive learning. Why is the example a data-design problem rather than a mistake in the cross-entropy derivative?
8. A model obtains a high linear-probe score on object labels. State two claims this supports and three claims it does not support about a representation used by a world model.
9. Explain why passive future-latent retrieval is not evidence that a learned model can predict the consequence of a chosen action.

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